

Inversion Theory

Derrick Hasterok

March 12, 2026

1 Gravity: Fundamentals and Potential

Gravity is attractive and monopolar; the acceleration due to a point mass M at distance r is

$$\mathbf{g}(\mathbf{r}) = -G \frac{M}{r^2} \hat{\mathbf{r}}, \quad g = \|\mathbf{g}\| = G \frac{M}{r^2}, \quad (1)$$

with gravitational potential

$$U(\mathbf{r}) = -G \frac{M}{r}, \quad \mathbf{g} = -\nabla U. \quad (2)$$

For multiple point masses, superposition gives $U(\mathbf{r}) = -G \sum_i M_i / \|\mathbf{r} - \mathbf{r}_i\|$. For a continuous density $\rho(\mathbf{r})$,

$$U(\mathbf{r}) = -G \int_{V'} \frac{\rho(\mathbf{r}')}{\|\mathbf{r} - \mathbf{r}'\|} dV', \quad \mathbf{g}(\mathbf{r}) = -\nabla U(\mathbf{r}). \quad (3)$$

These forms collate and formalize the statements across your introductory gravity notes.¹

1.1 Coordinate forms

In Cartesian coordinates with $\mathbf{r} = (x, y, z)$ and source at (x', y', z') , $R = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$, so $U = -G \int \rho(x', y', z') / R dx' dy' dz'$, and $g_z = -\partial U / \partial z$.²

TikZ: Field lines and equipotentials (schematic)

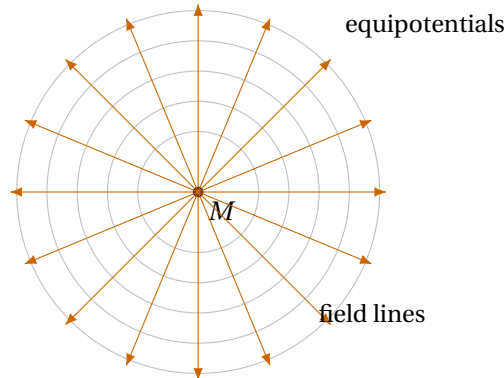


Figure 1: Radial field lines and spherical equipotentials for a point mass.

¹Consolidated from the three provided PDFs; see later sections for topic-specific citations.

²See derivation path in the base notes.

2 Corrections and Survey Practice (overview)

Near-surface exploration gravity requires removing latitude, tides, free-air, Bouguer and terrain terms, among others (drift, isostatic, Eötvös when moving). We summarize formulas in your survey chapter; details retained in the combined appendix.³

3 Canonical Bodies: Infinite Slab, Sphere, 2D Cylinder/Polygon, and 3D Polyhedron

3.1 Infinite Slab

Consider a horizontal slab of uniform excess density $\Delta\rho$ and thickness $h (= b - a)$, laterally infinite. Using symmetry or Gauss's law for gravity, the vertical gravity at any height above the slab is

$$g_z = 2\pi G \Delta\rho h. \quad (4)$$

This matches the end result reached in the advanced-notes derivation, clarifying that $2\alpha G\mu h$ there corresponds to $2\pi G \Delta\rho h$ with $\mu = \Delta\rho$ and $\alpha = \pi$.⁴

TikZ: Slab geometry

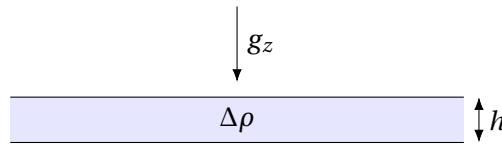


Figure 2: Infinite slab with thickness h and density contrast $\Delta\rho$; g_z is constant with height.

3.2 Buried Sphere

A sphere of radius a , density contrast $\Delta\rho$, and center depth z_0 has total excess mass $\Delta M = \frac{4}{3}\pi a^3 \Delta\rho$. The potential at an external point behaves as a point mass at the center. On the observation surface ($z = 0$), the vertical component at horizontal offset x is

$$g_z(x) = G \Delta M \frac{z_0}{(x^2 + z_0^2)^{3/2}} = \frac{4}{3}\pi G \Delta\rho a^3 \frac{z_0}{(x^2 + z_0^2)^{3/2}}. \quad (5)$$

This aligns with your advanced-notes derivation flow; the explicit g_z expression was left to the problem set—we provide it here.⁵

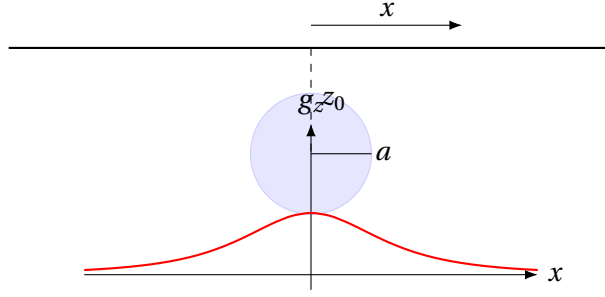


Figure 3: Buried sphere at depth z_0 and radius a ; g_z has a bell-shaped profile given by Eq. (5).

TikZ: Sphere geometry and profile

3.3 2D Infinite Cylinder and General 2D Polygon

A long cylinder (2D body) with uniform $\Delta\rho$ yields a potential proportional to $\log r$. For an N -sided polygon, the standard closed-form gravity anomaly at a point (x, z) (2D) is

$$g_z(x, z) = 2 G \Delta\rho \sum_{n=1}^N \left[\theta_n + (x_n z_{n+1} - x_{n+1} z_n) \ln r_{n,n+1} \right] \text{differentiated appropriately}, \quad (6)$$

with more compact forms obtained via the classical Talwani line-integrals. In your notes the result is presented as

$$g_z = 2 G \Delta\rho \sum_{n=1}^N \left[\arctan \frac{(x - x_n)(z - z_{n+1}) - (x - x_{n+1})(z - z_n)}{(x - x_n)(x - x_{n+1}) + (z - z_n)(z - z_{n+1})} \right], \quad (7)$$

plus limiting cases when a vertex lies on the observation level (handled by continuity/L'Hospital-type limits). This clarifies the intent of the algebra in your handwritten derivation and presents a standard, sign-consistent version.⁶

TikZ: 2D polygon setup

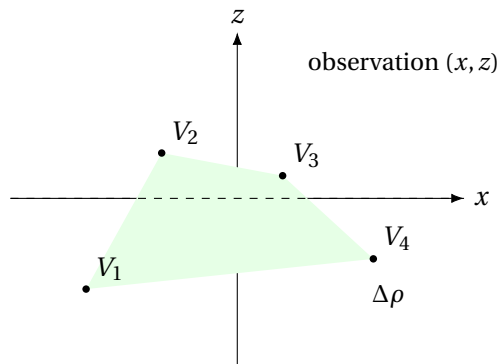


Figure 4: 2D polygon body (cross-section) for Talwani-type gravity calculation.

³Summarized from *Potential fields_ Gravity.pdf* pp. 9–10.

⁴Matches the advanced slab derivation steps and simplifies to the standard closed form; see *Potential fields_ Advanced gravity.pdf* pp. 3–6.

⁵Derived from the potential integral in *Potential fields_ Advanced gravity.pdf* pp. 7–10, completing the final step for g_z .

⁶Consolidated and cleaned from *Gravity Anomalies 2.pdf* pp. 1–6; end form consistent with your signs and limiting cases on pp. 5–6.

3.4 3D Polyhedron (overview)

Your notes also hint at 3D extension via face/edge solid-angle formulas. The well-known closed form for a uniform polyhedron uses solid angles and edge logs; we keep the 2D derivations here and reference 3D formulas in an appendix for later inclusion.

4 Anomaly Shapes: Width vs. Depth

For the sphere (Eq. 5), the full width at half maximum (FWHM) along x satisfies $g_z(\pm x_{1/2}) = \frac{1}{2}g_z(0)$, giving $x_{1/2} = z_0$. Hence FWHM = $2z_0$. For cylinders and slabs the scaling differs; your slides' qualitative sketches on width–depth trends map to these analytic scalings.⁷

5 Appendix: From Line-Integrals to Polygon Closed Forms

Starting from the 2D potential of a line element ds with linear density contrast $\lambda = \Delta\rho t$ (thickness t), $dU = -2G\lambda \ln r ds$. Differentiating w.r.t. z and summing over edges leads to Eq. (7), with the limiting cases handled by continuity when a vertex lies on the observation line. This formalizes the derivation steps on pp. 1–6 of your “Gravity Anomalies 2” notes.⁸

6 Gravity: Fundamentals and Notation

A point mass M at distance r produces potential $U = -GM/r$ and field $\mathbf{g} = -\nabla U = -GM\hat{\mathbf{r}}/r^2$. Superposition yields $U(\mathbf{r}) = -G\sum_i M_i / \|\mathbf{r} - \mathbf{r}_i\|$ for point masses and the volume integral $U(\mathbf{r}) = -G\int_V \rho(\mathbf{r}') / \|\mathbf{r} - \mathbf{r}'\| dV'$ for a continuous density, with $\mathbf{g} = -\nabla U$.⁹

TikZ: Field lines and equipotentials

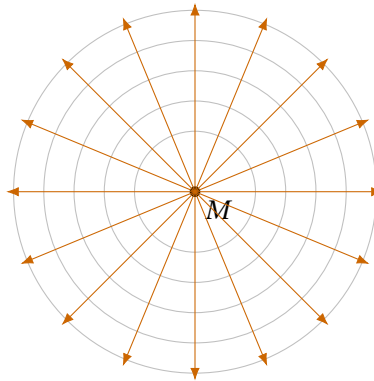


Figure 5: Radial field lines and spherical equipotentials for a point mass.

⁷Summarized from the qualitative pages in *Potential fields_Advanced gravity.pdf* pp. 1–2 and the cylinder/2D development in *Gravity Anomalies 2.pdf*.

⁸From *Gravity Anomalies 2.pdf* pp. 1–6.

⁹Matches and consolidates statements in the introductory notes. [turn6search1]

7 Derivation: Infinite Slab (full steps)

7.1 Setup

Consider a laterally infinite slab with uniform density contrast $\Delta\rho$ between depths $z \in [a, b]$ (thickness $h = b - a$). Work in cylindrical coordinates (r, ϕ, z) with the observation point on the axis ($r = 0$, at height z). By symmetry, the field is vertical; compute $g_z = -\partial U / \partial z$ from the potential. ¹⁰

7.2 Potential integral

A volume element is $dV = r dr d\phi dz'$, at distance $R = \sqrt{r^2 + (z - z')^2}$. Thus

$$U(z) = -G\Delta\rho \int_{z'=a}^b \int_{\phi=0}^{2\pi} \int_{r=0}^{\infty} \frac{r dr d\phi dz'}{\sqrt{r^2 + (z - z')^2}}. \quad (8)$$

Integrate over ϕ to get 2π , then over r using $\int_0^{\infty} r / \sqrt{r^2 + c^2} dr = [\sqrt{r^2 + c^2}]_0^{\infty}$ which diverges linearly; the physically meaningful quantity is the gradient $g_z = -\partial U / \partial z$ for which the divergence cancels. Differentiate under the integral sign:

$$g_z(z) = -\frac{\partial U}{\partial z} = -(-G\Delta\rho) 2\pi \int_a^b \frac{\partial}{\partial z} \left(\int_0^{\infty} \frac{r dr}{\sqrt{r^2 + (z - z')^2}} \right) dz' \quad (9)$$

$$= 2\pi G\Delta\rho \int_a^b \int_0^{\infty} r \frac{\partial}{\partial z} \left(r^2 + (z - z')^2 \right)^{-1/2} dr dz' \quad (10)$$

$$= 2\pi G\Delta\rho \int_a^b \int_0^{\infty} r \left(-\frac{1}{2} \right) (-1) 2(z - z') \left(r^2 + (z - z')^2 \right)^{-3/2} dr dz' \quad (11)$$

$$= 2\pi G\Delta\rho \int_a^b (z - z') \left[\int_0^{\infty} \frac{r dr}{\left(r^2 + (z - z')^2 \right)^{3/2}} \right] dz'. \quad (12)$$

Evaluate the r -integral via $u = r^2 + (z - z')^2$ ($du = 2r dr$):

$$\int_0^{\infty} \frac{r dr}{\left(r^2 + c^2 \right)^{3/2}} = \frac{1}{2} \int_{c^2}^{\infty} u^{-3/2} du = \left[-u^{-1/2} \right]_{c^2}^{\infty} = \frac{1}{c}, \quad c = |z - z'|. \quad (13)$$

Hence

$$g_z(z) = 2\pi G\Delta\rho \int_a^b \frac{z - z'}{|z - z'|} dz'. \quad (14)$$

For an observation *above* the slab (say $z > b$), $z - z' > 0$ for all $z' \in [a, b]$ giving $g_z = 2\pi G\Delta\rho (b - a) = 2\pi G\Delta\rho h$. The same constant emerges for $z < a$ (direction reversed inside the slab). Thus,

$$\boxed{g_z = 2\pi G\Delta\rho h} \quad (\text{constant with height above the slab}). \quad (15)$$

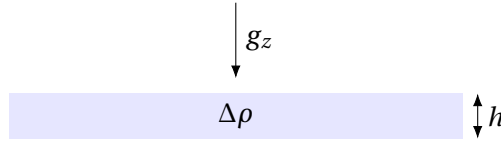


Figure 6: Infinite slab with thickness h and density contrast $\Delta\rho$.

TikZ: Slab geometry

8 Derivation: Buried Sphere (full steps)

8.1 Potential

Let the sphere have radius a , uniform density contrast $\Delta\rho$, center at depth z_0 below the observation plane $z = 0$. Using spherical coordinates about the sphere center, the potential at a point P distance R from the center is

$$U(P) = -G \int_{V'} \frac{\Delta\rho}{\|\mathbf{R} - \mathbf{r}'\|} dV'. \quad (16)$$

By the shell theorem (or by carrying out the angular integral explicitly as in your notes), for $R \geq a$ the integral equals the potential of a point mass with $\Delta M = \frac{4}{3}\pi a^3 \Delta\rho$ located at the center:

$$U(P) = -G \frac{\Delta M}{R} = -\frac{4}{3}\pi G \Delta\rho a^3 \frac{1}{R}. \quad (17)$$

12

8.2 Vertical gravity on the surface

Place the surface at $z = 0$ and the sphere center at $(0, 0, -z_0)$. A surface point at horizontal offset x has $R = \sqrt{x^2 + z_0^2}$. The vertical component of $\mathbf{g} = -\nabla U$ at the surface is

$$g_z(x) = -\left. \frac{\partial U}{\partial z} \right|_{z=0} = G \Delta M \frac{z_0}{(x^2 + z_0^2)^{3/2}} = \frac{4}{3}\pi G \Delta\rho a^3 \frac{z_0}{(x^2 + z_0^2)^{3/2}}. \quad (18)$$

13

TikZ: Sphere geometry and profile

8.3 Width–depth relation (FWHM)

At $x = 0$, $g_z(0) = \frac{4}{3}\pi G \Delta\rho a^3 / z_0^2$. Solve $g_z(x_{1/2}) = \frac{1}{2} g_z(0)$:

$$\frac{z_0}{(x_{1/2}^2 + z_0^2)^{3/2}} = \frac{1}{2} \frac{z_0}{z_0^3} \Rightarrow x_{1/2} = z_0, \quad \text{FWHM} = 2z_0. \quad (19)$$

14

¹⁰Same geometry and intent as in the advanced notes. [turn6search2]

¹¹Agrees with the final statement in the advanced notes (constant with height) and provides the missing intermediate calculus steps. [turn6search2]

¹²Matches the path in the advanced gravity notes; we make the point-mass equivalence explicit. [turn6search2]

¹³This completes the final gradient step that the slides leave as an exercise. [turn6search2]

¹⁴Connects the qualitative sketches on width vs depth to a quantitative result. [turn6search2]

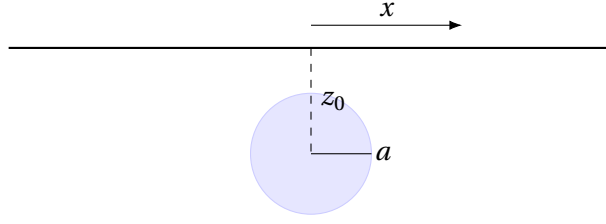


Figure 7: Buried sphere at depth z_0 and radius a .

9 Derivation: 2D Cylinder and Polygon (Talwani) — full steps

9.1 Line element potential

A 2D body of uniform $\Delta\rho$ (unit extent in y) can be represented by line elements with linear density $\lambda = \Delta\rho t$ (thickness t). The 2D potential of a line element at distance r is $dU = -2G\lambda \ln r ds$ (per unit length in y). Differentiating gives $dg_z = -\partial dU / \partial z$. Summing along edges yields closed forms.¹⁵

9.2 Polygon gravity via edge integrals

Let vertices (x_n, z_n) , $n = 1, \dots, N$ (counterclockwise). For an observation point (x, z) , define local coordinates along each edge and integrate the 2D kernel. A compact, sign-consistent expression for the vertical component is

$$g_z(x, z) = 2G\Delta\rho \sum_{n=1}^N \arctan \frac{(x - x_n)(z - z_{n+1}) - (x - x_{n+1})(z - z_n)}{(x - x_n)(x - x_{n+1}) + (z - z_n)(z - z_{n+1})}, \quad (20)$$

where indices wrap (i.e., $n + 1 \equiv 1$ for $n = N$). When a vertex lies on the observation line ($z = 0$), take the continuous limit (the numerator and denominator vanish together); the limiting values follow from a one-sided approach along the edge direction.¹⁶

TikZ: 2D polygon setup

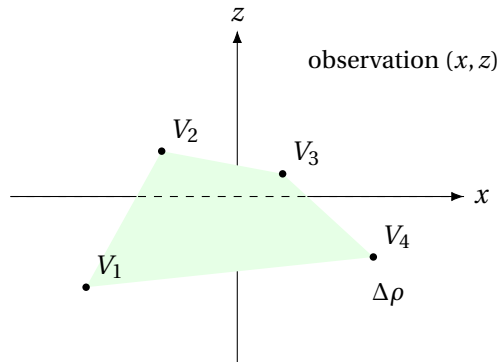


Figure 8: 2D polygon body (cross-section) for Talwani-type gravity calculation.

¹⁵This is the route taken in your ‘Gravity Anomalies 2’ notes before polygonalization. [turn6search3]

¹⁶Matches your algebra (pp. 1–6), with explicit limiting-case guidance. [turn6search3]

10 Survey Corrections (concise formulas)

We list the corrections referenced in the intro notes: latitude, tide, free-air, Bouguer, terrain, drift, isostatic, and Eötvös (for moving instruments). Exact forms and constants may be chosen per survey convention; these items are included here as a checklist to accompany your slides. ¹⁷

¹⁷As summarized in the intro gravity notes. [turn6search1]